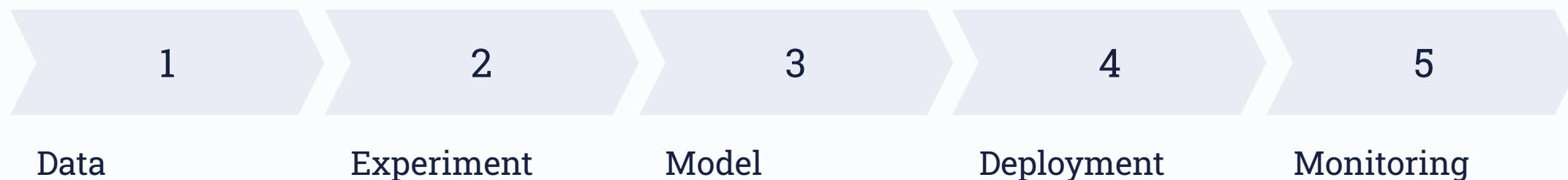


Introduction to MLflow

An Open-Source Platform for Managing the ML Lifecycle



MLflow helps manage machine learning experiments and models, connects experimentation with production workflows, and supports Data Scientists, ML Engineers, MLOps, and AI/Prompt Engineers.

Why Do We Need MLflow?

Challenges in ML Projects

- Many experiments and model versions
- Different parameters and hyperparameters
- Difficulty comparing experiment results
- Reproducing previous experiments can be difficult
- Managing models from development to production



MLflow provides a structured approach to managing these activities.

Core Components of MLflow

1

MLflow Tracking

Records parameters, metrics, and experiment runs

2

MLflow Projects

Packages ML code in a reproducible format

3

MLflow Models

Standardizes model packaging and deployment

4

MLflow Model Registry

Manages model versions and their lifecycle

Who Uses MLflow & How?



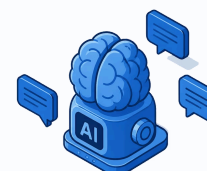
Data Scientists

- Experiment tracking
- Hyperparameter tuning
- Model comparison
- Code structuring & model packaging



MLOps / ML Engineers

- Model lifecycle & versioning
- Deployment dependencies
- Production deployment



Prompt / AI Engineers

- LLM & prompt experiments
- Model comparison
- Base model selection



Central Idea: MLflow connects experimentation with the production lifecycle.

MLflow: Complete Picture

Who Uses It & Why?

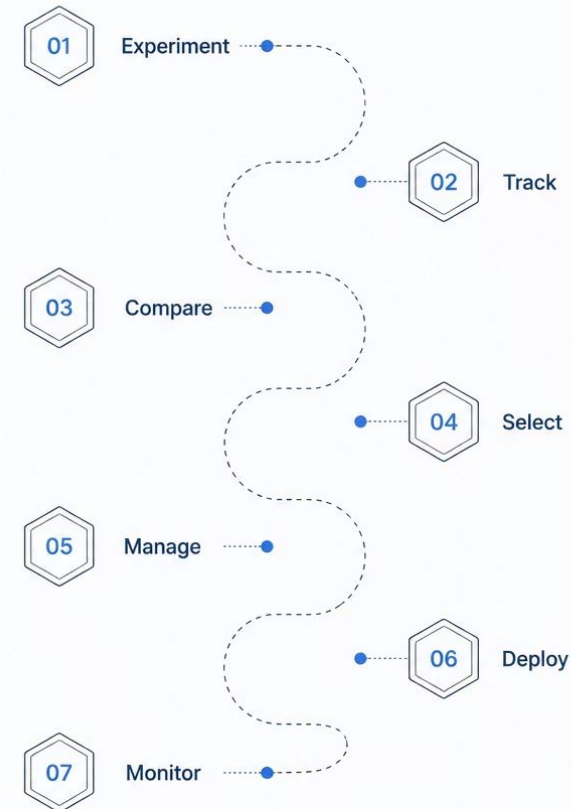
Data Scientist → Experiments, tuning, comparison

ML Engineer / MLOps → Model lifecycle, packaging, deployment

AI / Prompt Engineer → LLM and prompt experiments

Final Takeaway

MLflow brings structure, reproducibility, and lifecycle management to machine learning projects.



Topics We Will Cover

01

MLflow Tracking Server

03

MLflow UI

05

Creating Experiments & Runs

07

Comparing Experiments

09

Tracking ML Models & Artifacts

02

Setting Up MLflow

04

Setting Tracking URI

06

Logging Metrics

08

Visualizing Experiment Results

10

MLflow in End-to-End ML Projects